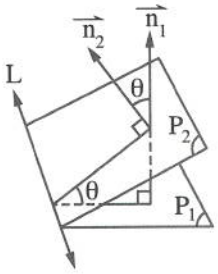
Unit 4

Lesson Three

Relative positions of two planes in space

If $P_1 : a_1 x + b_1 y + c_1 z + d_1 = 0$, $P_2 : a_2 x + b_2 y + c_2 z + d_2 = 0$ are equations of two different planes in space , then $\vec{n_1} = (a_1 , b_1 , c_1)$ is normal direction vector to the plane P_1 , $\vec{n_2} = (a_2 , b_2 , c_2)$ is normal direction vector to the plane P_2

The angle between two planes :



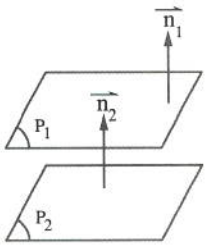
Let the measure of the smaller angle between the two planes P_1 , P_2 is θ where $\vec{n_1}$, $\vec{n_2}$ are the two normal direction vectors to them

∴ The measure of the angle between the two plane is given by the relation :

$$\cos\theta = \frac{|\vec{n_1} \cdot \vec{n_2}|}{\|\vec{n_1}\| \, \|\vec{n_2}\|} \quad \text{where } 0 \le \theta \le 90°$$

Two parallel planes and two perpendicular planes

1. Condition of two planes to be parallel ($P_1 \cap P_2 = \varnothing$) is : $\vec{n_1} \,//\, \vec{n_2}$

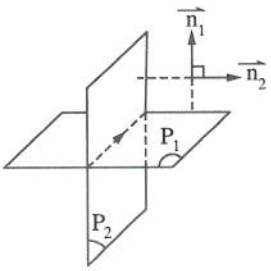


i.e. • $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

• $\vec{n_1} = k\,\vec{n_2}$

• $\vec{n_1} \times \vec{n_2} = \vec{o}$

2. Condition of two planes to be perperndicular ($\theta = 90°$) is $\vec{n_1} \cdot \vec{n_2} = 0$



i.e. $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

Remark

In the two previous planes P_1 , P_2 :

1. If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \neq \frac{d_1}{d_2}$, then the two planes are parallel and not coincident.

2. If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{d_1}{d_2}$, then the two planes are coincident.

Example 1

Find measure of the angle between the two planes :

$x + 2y - 3z = 20$ and $3x - 2y + 2z = 30$

Solution

$\overrightarrow{n_1} = (1 , 2 , -3)$, $\overrightarrow{n_2} = (3 , -2 , 2)$

$\therefore \cos\theta = \frac{|(1 , 2 , -3) . (3 , -2 , 2)|}{\sqrt{1+4+9}\ \sqrt{9+4+4}} = \frac{\sqrt{238}}{34}$ $\therefore \theta \simeq 63° \hat{1}$

Example 2

Find k that makes the two planes :

$3x - 4y + 2z + 9 = 0$, $3x + 4y - kz + 7 = 0$ are perpendicular.

Solution

$\overrightarrow{n_1} = (3 , -4 , 2)$, $\overrightarrow{n_2} = (3 , 4 , -k)$

, the two planes are perpendicular at $\overrightarrow{n_1} . \overrightarrow{n_2} = 0$

i.e. $(3 , -4 , 2) . (3 , 4 , -k) = 0$ $\therefore 9 - 16 - 2k = 0$ $\therefore k = -3.5$

Example 3

If measure of the angle between the two planes :

$3x - 6y + mz - 4 = 0$, $x + z - 7 = 0$ is 45° find m

Solution

$\overrightarrow{n_1} = (3 , -6 , m)$, $\overrightarrow{n_2} = (1 , 0 , 1)$

, $\because \cos\theta = \frac{|\overrightarrow{n_1} . \overrightarrow{n_2}|}{\|\overrightarrow{n_1}\| \|\overrightarrow{n_2}\|}$ $\therefore \frac{1}{\sqrt{2}} = \frac{|(3 , -6 , m) . (1 , 0 , 1)|}{\sqrt{9+36+m^2}\ \sqrt{1+1}}$

$\therefore \sqrt{45 + m^2} = |3 + m|$ "squaring the two sides"

$\therefore 45 + m^2 = 9 + 6m + m^2$ $\therefore 36 = 6m$ $\therefore m = 6$

The equation of the line of intersection of two planes

If $P_1 : a_1 x + b_1 y + c_1 z + d_1 = 0$, $P_2 : a_2 x + b_2 y + c_2 z + d_2 = 0$

are equation of two different planes in space and the ratios $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$ are not all equal, then the two planes are intersecting and we can find the equation of the intersection line by several methods, the following example illustrates some of them.

Example 4

Find the equation of the intersection line of the two planes :

$P_1 : x + y + 2z + 1 = 0$, $P_2 : 2x + y - z + 1 = 0$

Solution

Equations of the two planes are $x + y + 2z + 1 = 0$ (1)

and $2x + y - z + 1 = 0$ (2)

, $\because \frac{1}{2} \neq \frac{1}{1}$ $\therefore$ The two planes are intersecting

, subtracting equation (1) from equation (2) (to omit the variable y)

$\therefore x - 3z = 0$ $\therefore x = 3z$ (3)

Multiplying equation (2) by 2 and add equation (1) (to omit variable z)

$\therefore 5x + 3y + 3 = 0$ $\therefore x = \frac{-3-3y}{5}$ (4)

From (3) , (4) : $\therefore$ equation of the intersection line is : $x = \frac{-3-3y}{5} = 3z$

i.e. $\frac{x}{1} = \frac{y+1}{\left(-\frac{5}{3}\right)} = \frac{z}{\left(\frac{1}{3}\right)}$ **i.e.** $\frac{x}{-3} = \frac{y+1}{5} = \frac{z}{-1}$ (cartesian form)

Another solution

$x + y + 2z + 1 = 0$ (1) , $2x + y - z + 1 = 0$ (2)

Subtracting (1) from (2) : $\therefore x - 3z = 0$

, let $z = t$ $\therefore x = 3t$, substituting in (1) :

$\therefore 3t + y + 2t + 1 = 0$ $\therefore y = -1 - 5t$

$\therefore$ Parametrics equations of the line of intersection line are :

$\therefore x = 3t$, $y = -1 - 5t$, $z = t$

$\therefore \frac{x}{3} = \frac{y+1}{-5} = \frac{z}{1}$ **i.e.** $\frac{x}{-3} = \frac{y+1}{5} = \frac{z}{-1}$ (cartesian form)

Third solution

$\because$ Line of intersection of the two planes is perpendicular to both the normal direction vectors of the two planes which are $\overrightarrow{n_1}$, $\overrightarrow{n_2}$

, $\because \vec{n_1} = (1 , 1 , 2)$, $\vec{n_2} = (2 , 1 , -1)$

$\therefore \vec{d} = (\vec{n_1} \times \vec{n_2})$ is direction vector of the intersecting line

$$\therefore \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -3\hat{i} + 5\hat{j} - \hat{k}$$

to find a point belongs to the line of intersection we put $x = 2$ "or any other number" in the equations of the two planes

$\therefore y + 2z = -3$ (1) , $y - z = -5$ (2)

Solving (1) , (2) : $\therefore z = \frac{2}{3}$, $y = \frac{-13}{3}$

$\therefore$ The point $\left(2 , \frac{-13}{3} , \frac{2}{3}\right)$ lies on the line of intersection

$\therefore$ Vector form of the line of intersection is :

$$\vec{r} = \left(2 , \frac{-13}{3} , \frac{2}{3}\right) + t(-3 , 5 , -1)$$

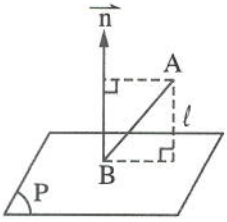
$$\therefore \frac{x-2}{-3} = \frac{y + \frac{13}{3}}{5} = \frac{z - \frac{2}{3}}{-1}$$ (cartesian form)

Notice that

You can find two points on the intersection line and so you can find the direction vector of the intersection line.

Length of the perpendicular drawn from a point to a plane

If $A(x_1 , y_1 , z_1) \notin$ plane P where $P : ax + by + cz + d = 0$ at which $\vec{n} = (a , b , c)$ is a normal direction vector , $B(x_2 , y_2 , z_2)$ any point lie on the plane P , then the length of the drawn perpendicular from the point A to the plane P which is denoted by ℓ is the length of the projection of $\overrightarrow{BA}$ in direction of $\vec{n}$



i.e. $$\ell = \frac{|\overrightarrow{BA} \cdot \vec{n}|}{\|\vec{n}\|}$$

, $\because \overrightarrow{BA} = \vec{A} - B = (x_1 , y_1 , z_1) - (x_2 , y_2 , z_2) = (x_1 - x_2 , y_1 - y_2 , z_1 - z_2)$

$$\therefore \ell = \frac{|(x_1 - x_2 , y_1 - y_2 , z_1 - z_2) \cdot (a , b , c)|}{\sqrt{a^2 + b^2 + c^2}}$$

$$\therefore \ell = \frac{|a(x_1 - x_2) + b(y_1 - y_2) + c(z_1 - z_2)|}{\sqrt{a^2 + b^2 + c^2}}$$

$$\therefore \ell = \frac{|ax_1 + by_1 + cz_1 - (ax_2 + by_2 + cz_2)|}{\sqrt{a^2 + b^2 + c^2}}$$

, $\because$ B $(x_2 , y_2 , z_2) \in$ the plane P

$\therefore a x_2 + b y_2 + c z_2 = - d$ (from the plane equation)

$$\therefore \boxed{\ell = \frac{|a x_1 + b y_1 + c z_1 + d|}{\sqrt{a^2 + b^2 + c^2}}}$$

Which is the cartesian form of the length of the perpendicular drawn from a point to the plane

Example 5

Find length of the perpendicular drawn from the point A (3 , 5 , 1) to the plane whose equation :

1. $x + 2y - 2z + 5 = 0$ **2.** $\vec{r} . (1 , 2 , - 1) = 2$

Solution

1. $\because$ Equation of the plane is $x + 2y - 2z + 5 = 0$

, the point is A (3 , 5 , 1)

$$\therefore \text{Length of the perpendicular} = \frac{|3 + 2(5) - 2(1) + 5|}{\sqrt{1 + 4 + 4}}$$

$$= \frac{|16|}{3} = 5\frac{1}{3} \text{ length unit.}$$

2. $\because \vec{r} . (1 , 2 , - 1) = 2$ $\therefore x + 2y - z - 2 = 0$

, $\because$ the point is A (3 , 5 , 1)

$$\therefore \text{Length of the perpendicular } \ell = \frac{|3 + 2(5) - (1) - 2|}{\sqrt{1 + 4 + 1}} = \frac{|10|}{\sqrt{6}} = \frac{5\sqrt{6}}{3} \text{ length unit.}$$

The distance between two parallel planes

To find the distance between two parallel planes in space, find a point on one of them , then find the length of the perpendicular from this point to the other plane.

Example 6

Prove that , the two planes : $P_1 : x - 2y - 2z - 12 = 0$, $P_2 : \vec{r} . (- 3 , 6 , 6) = 9$ are parallel , then find the distance between them

Solution

For the first plane $\vec{n_1} = (1 , - 2 , - 2)$

, for the second plane $\vec{n_2} = (- 3 , 6 , 6)$

, $\because \frac{1}{-3} = \frac{-2}{6} = \frac{-2}{6}$

$\therefore$ The two planes are parallel

, putting $x = -1$, $y = -2$

in the equation of P_2

$\therefore (-1, -2, z) . (-3, 6, 6) = 9$

$\therefore 3 - 12 + 6z = 9$

$\therefore z = 3$

$\therefore$ The point $(-1, -2, 3) \in$ the plane P_2

$\therefore$ The distance between the two planes = length of the perpendicular drawn from $(-1, -2, 3)$ to P_1

$\therefore \ell = \frac{|-1-2(-2)-2(3)-12|}{\sqrt{1+4+4}} = 5$ length unit.

Notice that

To find a point B $\in$ the plane, put two arbitrary values for x, y and calculate the corresponding value of z

Also, you can choose two values of x and z, then calculate the corresponding value of y or choose two values of z and y, then calculate the corresponding value of x

Example 7

Find the equation of the plane parallel to the plane : $2x + y - 4z + 5 = 0$ and at a distance $\sqrt{21}$ length unit from the point $(1, 2, 0)$

Solution

$\because$ The required plane is parallel to the plane : $2x + y - 4z + 5 = 0$

$\therefore$ Its equation is : $2x + y - 4z + d = 0$

, $\because$ the length of the perpendicular drawn from $(1, 2, 0)$ to the required plane $= \sqrt{21}$ length unit

$\therefore \sqrt{21} = \frac{|2(1)+(2)-4(0)+d|}{\sqrt{4+1+16}}$ $\quad \therefore 21 = |4 + d|$

$\therefore 4 + d = 21$ $\quad \therefore d = 17$

Or $4 + d = -21$ $\quad \therefore d = -25$

$\therefore$ The two planes $2x + y - 4z + 17 = 0$, $2x + y - 4z - 25 = 0$ are parallel to the plane $2x + y - 4z + 5 = 0$ and each of them at a distance $\sqrt{21}$ length unit from the point $(1, 2, 0)$

Example 8

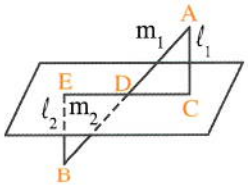
Find the ratio by which the plane : $2x + 3y - z = 5$ divides the distance between the two points A (1 , 1 , 1) and B (3 , 2 , – 1)

Solution

$\because 2(1) + 3(1) - (1) - 5 = -1 < 0$, $\because 2(3) + 3(2) + 1 - 5 = 8 > 0$

$\therefore$ A , B are on different sides of the plane

$\therefore$ The plane divide $\overline{AB}$ internally by the ratio $m_1 : m_2$,



from similarity of $\Delta\Delta$ ACD , BED , then $\frac{m_1}{m_2} = \frac{\ell_1}{\ell_2}$

Where ℓ_1 , ℓ_2 are the lengths of the perpendicular drawn from A , B to the plane respectively

, $\because \ell_1 = \frac{|2(1) + 3(1) - (1) - 5|}{\sqrt{4+9+1}} = \frac{1}{\sqrt{14}}$, $\ell_2 = \frac{|2(3) + 3(2) - (-1) - 5|}{\sqrt{4+9+1}} = \frac{8}{\sqrt{14}}$

$\therefore \ell_1 : \ell_2 = 1 : 8$ $\therefore m_1 : m_2 = 1 : 8$

Remark

If $A = (x_1 , y_1 , z_1)$, $B = (x_2 , y_2 , z_2)$ and D divides $\overline{AB}$ internally by ratio $m_1 : m_2$, then the coordinates of $D = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} , \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} , \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2}\right)$

The previous example can be solved as follow :

Find the cooordinates of the division point

D which divides AB internally $= \left(\frac{3m_1 + m_2}{m_1 + m_2} , \frac{2m_1 + m_2}{m_1 + m_2} , \frac{-m_1 + m_2}{m_1 + m_2}\right)$

, $\because$ this point belongs to the plane

$\therefore 2\left(\frac{3m_1 + m_2}{m_1 + m_2}\right) + 3\left(\frac{2m_1 + m_2}{m_1 + m_2}\right) - \left(\frac{-m_1 + m_2}{m_1 + m_2}\right) = 5$

$\therefore 6m_1 + 2m_2 + 6m_1 + 3m_2 + m_1 - m_2 = 5m_1 + 5m_2$

$\therefore 8m_1 = m_2$ $\therefore \frac{m_1}{m_2} = \frac{1}{8}$